

A-level

# ENVIRONMENTAL SCIENCE

## Paper 1 - Generated practice paper Set 33

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <p><i>Up to 6 marks in total; award marks in proportion to the completeness of each point.</i></p> <ul style="list-style-type: none"> <li>• Detritivores (worms, beetle larvae, millipedes, woodlice) physically break up dead organic matter, increasing its surface area, and release nutrients; worms create tunnels improving drainage and aeration (<b>up to 2</b>)</li> <li>• Decomposers (bacteria, fungi) secrete enzymes that break down dead organic matter, releasing nutrient ions (ammonification) and forming humus which binds particles into crumb peds (<b>up to 2</b>)</li> <li>• Nitrogen-fixing bacteria convert nitrogen gas to ammonium ions; nitrifying bacteria oxidise ammonium to nitrite then nitrate for root absorption (<b>up to 2</b>)</li> <li>• Mycorrhizal fungi form symbiotic associations with roots increasing phosphate uptake (<b>up to 2</b>)</li> </ul> | 6     |
| 01.2     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• This is the range of tolerance of most plants and soil biota; acidic soils leach nutrients and mobilise toxic aluminium, damaging root membranes (<b>1</b>)</li> <li>• In alkaline soils phosphates become insoluble / unavailable (<b>1</b>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| 01.3     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Interstitial spaces are reduced so aeration falls (aerobic processes such as nitrification and decomposition slow) and roots cannot penetrate (<b>1</b>)</li> <li>• Infiltration falls so waterlogging is more likely and runoff / erosion increase (<b>1</b>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 158-161; AQA spec 3.2.5.1.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>02.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated <b>(1)</b></li> <li>• Grit traps: the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |
| <b>02.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge <b>(1)</b></li> <li>• This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2     |
| <b>02.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter <b>(1)</b></li> <li>• Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process <b>(1)</b></li> <li>• The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge'; returning them maintains a large bacterial population so digestion is rapid <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                        | 3     |
| <b>02.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria <b>(1)</b></li> <li>• Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) <b>(1)</b></li> <li>• Phosphate stripping with iron(III) sulfate; micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                              | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
| <b>03.1</b> | <p><i>Any 4 from the points below - 1 mark each, maximum 4.</i></p> <ul style="list-style-type: none"> <li>• Topography: a narrow exit from a large deep basin so a small dam holds a large volume <b>(1)</b></li> <li>• Geology: impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity <b>(1)</b></li> <li>• Catchment area: large enough with regular, reliable rainfall / river inflow; climate not too hot (evaporation losses) <b>(1)</b></li> <li>• Catchment land use: low pollution risk (industry, pesticides); low soil erosion (sedimentation reduces capacity) <b>(1)</b></li> <li>• Infrastructure / proximity to demand: access for construction, cost of transporting water <b>(1)</b></li> <li>• Existing land use: value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis <b>(1)</b></li> </ul> | 4     |
| <b>03.2</b> | <p><i>Up to 3 marks in total; award marks in proportion to the completeness of each point.</i></p> <ul style="list-style-type: none"> <li>• Advantage: fewer land-use conflicts as no land is flooded; large volume near coastal cities <b>(up to 2)</b></li> <li>• Disadvantages: destroys / changes important intertidal habitats (mudflats for wading birds); obstacle to shipping; collects pollution from the whole river catchment; sedimentation <b>(up to 2)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 | 3     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>03.3</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Very energy-intensive and expensive (reverse osmosis / distillation) <b>(1)</b></li> <li>• Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| <b>03.4</b> | <p>1 mark.</p> <ul style="list-style-type: none"> <li>• Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
| <b>04.1</b> | <p>2 marks for each point (partial credit for a partly correct point).</p> <ul style="list-style-type: none"> <li>• Ground surface: pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals; trapped predators eat others; preservative attracts / repels; escapes <b>(2)</b></li> <li>• Leaf litter: Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light; some die or stop moving; large ones cannot pass the mesh <b>(2)</b></li> <li>• Branches: beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour; some species not dislodged / fly away; high branches unreachable <b>(2)</b></li> </ul> | 6     |
| <b>04.2</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught <b>(1)</b></li> <li>• Brightness / wavelength of the lamp affects attractiveness so catches vary between traps <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
| <b>04.3</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• The area of ground covered by the suction pipe is known so invertebrates per m<sup>2</sup> can be calculated <b>(1)</b></li> <li>• A pooter is only a handling / collection device for individuals already found, not a sampling method <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |
| <b>05.1</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• The human ear is sensitive to a very wide range of sound pressures / powers (about 10<sup>12</sup> range) <b>(1)</b></li> <li>• A logarithmic scale compresses this into a convenient range of numbers: each 10 dB increase represents a ten-fold increase in sound intensity <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| <b>05.2</b> | <p>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</p> <ul style="list-style-type: none"> <li>• Difference = 30 dB = 3 x 10 dB <b>(1)</b></li> <li>• = 10<sup>3</sup> = 1000 times <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |
| <b>05.3</b> | <p>1 mark for each point.</p> <ul style="list-style-type: none"> <li>• 88 dB <b>(1)</b></li> <li>• Doubling the sound intensity increases the level by 3 dB (logarithmic scale) <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2     |
| <b>05.4</b> | <p>1 mark.</p> <ul style="list-style-type: none"> <li>• 0 dB is set as the quietest sound that can be heard (at 1000 Hz) - the threshold of human hearing; quieter sounds exist but cannot be heard <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>05.5</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• 20 to 20 000 Hz <b>(1)</b></li> <li>• dB(A) weights sounds between about 1000 and 4000 Hz more heavily <b>(1)</b></li> <li>• Because the nerve cells / hair cells in the cochlea are more sensitive to these frequencies so they are heard more easily / cause more annoyance; range narrows with age or noise damage <b>(1)</b></li> </ul>                                                                                                                              | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-287; AQA spec 3.4.3.2.13.                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| <b>06.1</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• LEO: high resolution and coverage of the whole Earth (including the poles) allows detailed mapping of ice area / thickness and forest cover <b>(1)</b></li> <li>• But each area is only revisited every 15 days so they cannot follow rapidly changing weather <b>(1)</b></li> <li>• Geostationary: constant view of the same region allows continuous monitoring of cloud / storm movements, though the poles are not visible and detail is lower <b>(1)</b></li> </ul> | 3     |
| <b>06.2</b> | <p><i>Any 3 from the points below - 1 mark each, maximum 3.</i></p> <ul style="list-style-type: none"> <li>• Wind velocity <b>(1)</b></li> <li>• Ocean currents / sea surface temperature <b>(1)</b></li> <li>• Wave height <b>(1)</b></li> <li>• Ice thickness <b>(1)</b></li> <li>• Vegetation cover <b>(1)</b></li> <li>• Sea level <b>(1)</b></li> <li>• Atmospheric temperature / composition <b>(1)</b></li> </ul>                                                                                                                               | 3     |
| <b>06.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Data for a past year (eg 1900) are fed in and the model's prediction for a later year (eg 2000) is compared with the real outcome <b>(1)</b></li> <li>• If the prediction matches, the model can be trusted (with caution); it is continually modified as more data are collected and differences analysed <b>(1)</b></li> </ul>                                                                                                                                         | 2     |
| <b>06.4</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Models need data from all regions / oceans / altitudes; gaps (eg oceans, polar regions, past periods) introduce uncertainty into predictions <b>(1)</b></li> <li>• Unrepresentative data (eg urban heat islands, short time series) would bias the model; remote sensing helps fill gaps <b>(1)</b></li> </ul>                                                                                                                                                           | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 102; AQA spec 3.2.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |
| <b>07.1</b> | <p><i>Correct answer only.</i></p> <ul style="list-style-type: none"> <li>• C <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     |
| <b>07.2</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Monitoring can be focused on the sites predicted to be most at risk / where the pollutant will concentrate <b>(1)</b></li> <li>• No need to sample sites that were never likely to be polluted; sampling also checks the accuracy of the predictions <b>(1)</b></li> </ul>                                                                                                                                                                                               | 2     |
| <b>07.3</b> | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Behaviour of new chemicals can be predicted from similar existing chemicals: it is likely to persist, bioaccumulate and biomagnify <b>(1)</b></li> <li>• So acceptable emission limits, control methods and monitoring programmes can be set before release (precautionary principle) <b>(1)</b></li> </ul>                                                                                                                                                              | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243-245; AQA spec 3.4.3.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |
| <b>08.1</b> | <i>1 mark for the correct answer.</i><br>• $180 / 1.2 = 150$ <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1     |
| <b>08.2</b> | <i>1 mark.</i><br>• Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1     |
| <b>08.3</b> | <i>1 mark for each point.</i><br>• Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium ( $1.2$ vs $0.05 \text{ mg kg}^{-1}$ ) <b>(1)</b><br>• Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years <b>(1)</b><br>• Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease) <b>(1)</b><br>• Rice is the staple food so residents receive a large dose; village B has no exposure and no disease (correlation supports causation) <b>(1)</b> | 4     |
| <b>08.4</b> | <i>Any 2 from the points below - 1 mark each, maximum 2.</i><br>• Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble <b>(1)</b><br>• Cover / cap spoil heaps to prevent leaching and dust <b>(1)</b><br>• Recycle nickel-cadmium batteries / hazardous waste landfill <b>(1)</b><br>• Phytoremediation of contaminated paddy soils <b>(1)</b>                                                                                                                                                                                            | 2     |
| <b>08.5</b> | <i>1 mark for each point.</i><br>• Manufacture may release cadmium in effluents / dust <b>(1)</b><br>• Panels have a limited life; disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled <b>(1)</b>                                                                                                                                                                                                                                                                                                                                                   | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |
| <b>09.1</b> | <i>1 mark for each point.</i><br>• Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium <b>(1)</b><br>• Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet <b>(1)</b><br>• International agreement on ownership of the seabed and its resources is needed <b>(1)</b>                                                                                                                                                                  | 3     |
| <b>09.2</b> | <i>1 mark for each point.</i><br>• Collectors disturb the seabed and kill benthic organisms living on it <b>(1)</b><br>• Separating nodules from sediment creates plumes that increase turbidity with unknown consequences <b>(1)</b><br>• Re-deposition of sediment covers and kills filter feeders and other benthic organisms; slow recovery in the deep sea <b>(1)</b>                                                                                                                                                                                                                        | 3     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 09.3     | <p><i>1 mark for each point.</i></p> <ul style="list-style-type: none"> <li>• Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) <b>(1)</b></li> <li>• Exploitable land deposits are limited and mainly in China; chemical separation is complex and produces large amounts of toxic waste <b>(1)</b></li> <li>• Bacterial adsorption: dissolved rare earth ions adsorb onto the cell surfaces of bacteria (<i>Bacillus subtilis</i>) and are washed off at specific pHs to separate them <b>(1)</b></li> <li>• Deep sea sediments contain relatively high concentrations but no viable extraction method yet <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |
| 10.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Features: renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts</li> <li>• Fossil fuels: high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO<sub>2</sub>), politically concentrated (Middle East oil)</li> <li>• Nuclear: very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition</li> <li>• Renewables: low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs</li> <li>• Sustainability concepts: impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion)</li> <li>• Economic and political factors: subsidies, guaranteed prices, planning, external costs not included in price, energy security</li> <li>• Judgement: a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable; transition must start before depletion</li> </ul> | 9     |
| 10.2     | <p><i>1 mark.</i></p> <ul style="list-style-type: none"> <li>• Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station <b>(1)</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
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|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| 11.1     | <p><b>Discuss the extent to which the study of ozone depletion illustrates how scientific investigation can lead to the successful solution of an environmental problem.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Stages of investigation: identifying the issue (Halley Station UV data, Dobson unit measurements), hypothesis formation (Rowland-Molina 1974: persistence, UV dissociation, chlorine reactions), data collection (ground, satellite, balloon and aircraft air sampling detecting CIO), analysis of spatial, temporal and altitude variability, proposal of solutions, enactment</li> <li>• Why Antarctica: low stratospheric temperatures, ice crystals, stratospheric clouds, polar vortex, spring sunlight</li> <li>• Effects of UVB: skin cancer, cataracts, DNA damage, plant and marine damage - motivating action</li> <li>• Solutions: Vienna Convention, Montreal Protocol - phase-out and bans, alternative processes (pump sprays), alternative materials (HCFCs, HFCs, hydrocarbons, HFAs), collection and incineration of CFCs, fund for developing countries</li> <li>• Evaluation of success: ozone hole area / depth stabilising, CFC levels falling; limitations - persistence, remaining ODSs, HFCs as greenhouse gases, decades to full recovery</li> <li>• Comparison with other atmospheric problems (climate change, acid rain) <ul style="list-style-type: none"> <li>- why ozone was easier; role of peer review, international cooperation, availability of substitutes</li> </ul> </li> <li>• Conclusion on the extent to which it is a model for other issues</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 108-113; AQA spec 3.2.1.3.</p> | 25    |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks |
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| 11.2     | <p><b>Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use</li> <li>• Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles</li> <li>• Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour</li> <li>• Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management</li> <li>• Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour</li> <li>• Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.</p> | 25    |